

lec15

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1 Convolution

- Convolution is a type of information bottleneck
- Ex. Given $x_i = [0, 0.1, 0.2, 1.4, 1.5, 0.1, 0.2]$, we want the model to notice the sudden spike from $0.2 \rightarrow 1.4$
- Can apply convolution filter $f_1 = [-1, 1, 1, -1]$, calculate $F = f_1 * x_i, \forall i$
 - f_1 measures how x_k, x_{k+1} differ from x_{k-1}, x_{k+2}
- Then, take the sum of x_i 's elements with $f_2 = [1, 1, 1, 1, 1, 1]$

However, convolution only works if we impose a fixed size sliding window f , typically smaller than the original x_I .

2 Attention

- Let $x_i \in \mathbb{R}^d$ be an input vector, and let $q \in \mathbb{R}^d$ be an attention filter of the same dimension.
- Each component q_j controls how much attention is placed on the corresponding component x_{ij} of x_i :

$$x_i \odot q = \begin{bmatrix} x_{i1}q_1 \\ x_{i2}q_2 \\ \vdots \\ x_{id}q_d \end{bmatrix} \in \mathbb{R}^d$$

- A larger value of q_j means that the model places more weight on the j -th feature of x_i .

3 Large Language Models

- Given input C , an LLM predicts the next word w based on finding the posterior distribution $P(w|C)$.
- A LLM is trained on large amounts of data to predict the posterior
- Each word is assigned a unique ID, called a token t
- Then, this reduces to a signal processing problem - given a sequence of tokens $t_1 \dots t_k$, predict t_{k+1} .

Then, the training data for the next-word predictor takes on the following form:

$$x_1 = [t_1 \dots t_k], y_1 = t_{k+1}$$

We can then recursively run this:

First pass:

$$x_1 = [t_1 \dots t_k] \rightarrow \text{Predict } t_{k+1}$$

Second pass:

$$x_2 = [t_1, \dots, t_k, t_{k+1}] \rightarrow \text{Predict } t_{k+2}$$

and so on.

- Typically, training LLM take on the following form:
- **Foundational Model Training** - learn posterior from entire Internet
- **Mid-Training** - refine posterior on questions asked on the Internet - identifying a smaller subset of human data that looks like a conversation
- **Post-Training** - using real people to interact with the model, and finetune the model with sophisticated conversations